DP4b. Matched solves: per-mode sound radiation (acoustic transfer) of vibrating objects, each mode solved alone at its own frequency on a grid or surface matched to it, with radiation damping, air loading and an error certificate per mode

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Classification (CPC): G10K 15/02 (synthesis of acoustic waves); A63F 13/54 (game output involving acoustic signals); G06F 30/23 (simulation by finite element or finite difference methods); G06F 17/13 (differential equations); G06F 17/14 (Fourier transformations); G06F 17/16 (matrix or vector computation).

Related disclosures in this series: DP4 (sound made from the matter: where the modes come from and how they are played); DP5 (the physel primitive that objects are made of).

Derived means computed exactly from formulas given here; others' measurements are cited.

1. Field

Acoustics and computer simulation: computing, for every vibration mode of a solid object, the sound it radiates into the air and the damping and pitch change the air adds, fast enough for objects created while a game, virtual reality world or other simulation runs.

2. Problem and known methods

2.1 The problem

Modal sound synthesis plays a struck object as a sum of damped sine waves, one per vibration mode. What reaches a listener from each mode depends on how the mode radiates: its strength and phase in every direction, how fast radiation drains its energy (radiation damping), and how much the moving air lowers its pitch (air loading). Each needs a solution of the exterior Helmholtz equation at the mode's frequency, driven by the mode's motion on the object's outer surface.

An object can have hundreds to thousands of audible modes. At 29.4 kHz, near the top of hearing, the wavelength in air is 11.7 mm. When objects are made while the simulation runs (carved, built, broken), the solves should finish in seconds to minutes on one consumer GPU, in the background, and each result should say how accurate it is.

Damping and air loading matter for light objects. For a 1 mm aluminium plate, radiation can be 68-89% of a mode's decay, and air loading lowers the pitch by 0.72-1.5% between 200 Hz and 1 kHz; a 5 mm steel plate rings 13% shorter at 1 kHz and 35% shorter at 4 kHz because of radiation (derived from the infinite-plate formulas of [27]). Sound that aims at steps of 0.24 dB in level, 0.083% in pitch and 9.5-24% in

ringing time must compute all three.

2.2 Known methods

Method	What it does	Published figures (the authors')
Precomputed acoustic transfer, PAT [1]	One boundary-element (BEM) solve per mode; equivalent multipole sources fitted to each mode's field	A few tens of dipole sources a mode
FFAT maps [2]	Each mode's far-field transfer amplitude over directions as an image-like map with a short 1/r series, from fast multipole BEM	—
Interactive acoustic transfer approximation [3]	Prony series over frequency at key points; adaptive frequency sweeps; meshes simplified per frequency, keeping volume velocity	One Helmholtz solve: 4,647 s at 30,076 triangles, 370 s on a 2,004-triangle simplified mesh (20-core Xeon cluster); simplified-mesh L2 errors 0.2% at 832 Hz, 0.7% at 5.5 kHz, 8.1% at 12.3 kHz
KleinPAT [4]	Groups modes into "chords", runs one GPU time-domain simulation per chord, then separates each mode's transfer	292 modes of a model in 6 simulations; hundred- to thousand-fold speed-ups over Helmholtz-based solvers
Wave-based sound synthesis [5]; WaveBlender [6]	Sharp-interface FDTD wave solvers for animation sound; [6] on a uniform GPU grid	[6]: 0.73-86 times the sound's own length on an RTX 4090
NeuralSound [7]; NAT [8]	Networks predict modes and far-field maps [7], or encode transfer learned from BEM data [8]	[7]: under 1 s per new object on an RTX 3080 Ti
SonicRadiation [9]	FDTD far away coupled to a time-domain BEM near the object	—
Fast BEM products [10-14]	Fast multipole, hierarchical matrices, IFGF, precorrected FFT, AIM: operators in O(N) to O(N log N)	[10]: up to kD ≈ 500
Burton-Miller [15]; OSRC preconditioning [16, 17]	A combined equation without false resonances; an analytic preconditioner from an on-surface radiation condition	[17], with fast multipole products: GMRES needed 4-5 iterations on a sphere at k = 4.76-47.4 (10 points a wavelength) against 25-173 without the preconditioner, and 7 on a cone-sphere (k = 8, 8 points) against 176-235; far-field L2 errors 0.2-2.5%
Thin bodies [18]; equivalent sources [19]; low-frequency series [20, 21]; acoustic transfer vectors [22]	Hypersingular equations for sheets; sources inside the object; series in ka from Laplace problems; transfer to field points, often by reciprocity	—
Time-harmonic results from time-domain runs [23-26]	Controllability, WaveHoltz, FDTD phasor extraction, FDTD near-to-far-field transforms	—

2.3 What these leave open

1. **Cost that follows the top frequency.** A time-domain run must resolve the highest frequency it carries. On one grid sized for the top of hearing every mode costs as much as the top one, and that cost grows as the fourth power of the frequency (cells as its cube, time steps as its first power).
2. **Boundary solves per mode on CPUs** take minutes to hours at high frequency [3].
3. **Accuracy per result.** The methods above report accuracy from tests on sample shapes; we found none that attaches an error estimate, computed from the solve itself, to each mode's result.
4. **Damping and pitch without the air.** In the modal sound systems we found, decay comes

from material damping fitted to recordings (which include the recorded object's own radiation) and pitch from the modes in vacuum.
5. **Learned predictors** carry no error bound for a new shape.

3. Summary

Matched solves compute each mode's radiation on its own, at the mode's own frequency, on a discretization sized for that frequency rather than the highest one. Modes in the low band (ka below about 0.5) are solved at once on the CPU as a short series in frequency, all from one factored Laplace operator per shape (LB). Higher modes are solved either by a GPU time-domain run on a grid of 6-8 points a wavelength in a compact absorbing domain, with the steady state read by lock-in and the far field taken by an exact surface integral (MW, the matched wave solve), or by a frequency-domain solve on the outer surface only, with FFT-accelerated products and an OSRC-preconditioned Burton-Miller equation (MB, the matched boundary solve). Every mode yields complex multipole coefficients, its radiated power (so its radiation damping), its air loading and its coupling to near-equal modes, with a certificate estimating its error. Solves run loudest first in the background, and each mode switches once from an interim estimate to its exact result, at an audio block boundary.

4. Detailed description

4.1 Conventions and inputs

- Time factor $e^{(-i\omega t)}$; $k = \omega/c$; air at 20 °C: $c = 343$ m/s, $\rho_0 = 1.2$ kg/m³; $G(x, y) = e^{(ik|x-y|)}/(4\pi|x-y|)$.
- Γ is the object's outer, wetted surface, with unit normal n pointing into the air; x_0 is the expansion centre (the bounding box's centre); a is the radius of the smallest sphere about x_0 enclosing Γ; L is the largest side; $\lambda = c/f$.
- Per mode n: the in-vacuo angular frequency ω_n; the mass-normalized mode shape φ_n; its normal component $u_n = \varphi_n \cdot n$ on Γ; and a loudness weight w_n (the mode's expected share of energy, for ordering).
- The surface may be a mesh, a level set or a union of implicit primitives: anything that answers inside or outside, closest point and normal. With small implicit primitives (physels, DP5), Γ is the union of their exposed faces, faces between bonded primitives do not radiate, and sheet-like parts much thinner than the step are open sheets.
- The air moves with the surface, $\rho_0\, \partial v/\partial t = -\nabla p$, so for unit modal amplitude $\partial p/\partial n = q_n = \rho_0 \omega_n^2 u_n$ on Γ; p solves $\Delta p + k^2 p = 0$ outside Γ and radiates outward.

4.2 What each mode's result holds

Field	Type	Bytes
Mode index	u32	4
Frequency solved at (rad/s): in vacuo, or loaded after a second solve	f64	8
Part (LB, MW, MB), state (interim, exact), solver version	3 × u8	3
Multipole order N	u16	2
Expansion centre x_0 (object frame, m)	3 × f32	12
Coefficients a_{lm}, $l = 0 \ldots N$, $m = -l \ldots l$	complex f32	$8\,(N+1)^2$
Radiated power P (W at unit modal amplitude); radiation decay rate σ_rad (1/s); pitch shift $\Delta\omega/\omega$	3 × f32	12
Certificate: estimated relative far-field error (the sum of §4.8's parts, used by the pass test), its largest source, and MW's advisory dispersion bound	f32 + u8 + f32	9
Near-equal partners: count, then (index u32, Z_{mn} complex f32) each	u16 + 12 each	2 + 12 per partner

Outside the sphere of radius a, $p(x) = \Sigma_l \Sigma_m a_{lm}\, h_l(k|x - x_0|)\, Y_{lm}(\hat{x})$ (h_l the spherical Hankel function of the first kind, Y_{lm} orthonormal spherical harmonics). Results are a cache derived from the shape and its

modes, keyed by a digest of both and the solver version; identical copies share one result.

4.3 Which part solves which mode

Each mode gets $k_n a = \omega_n a/c$ (Figure 1). Below 0.5 (useful 0.3-0.8, always below 1): LB, on the CPU, all such modes at once. From 0.5 up: MW by default, on the GPU; MB for large objects, when its working memory fits what is free. One possible switch (an example): MB for an object when MW's cost estimate for it exceeds about 10 minutes at full GPU use.

4.4 LB, the low band

Write $q_n = \rho c^2 k^2 u_n$ and $p_n = \rho c^2 k^2 \tilde{p}$. The conventional boundary integral equation of the exterior Neumann problem is

$(\tfrac{1}{2}I - K_k)\,\tilde{p} = -V_k\,u_n$, $(K_k f)(x) = \int_\Gamma \partial G/\partial n_y\, f\, dS_y$, $(V_k f)(x) = \int_\Gamma G\, f\, dS_y$.

It has no false resonance here: the lowest one of any shape inside the sphere of radius a lies at $ka \geq \pi$. Expand $G = \Sigma_j (ik)^j r^{j-1}/(4\pi\, j!)$, $r = |x - y|$, so $K_k = \Sigma_j (ik)^j K_j$ and $V_k = \Sigma_j (ik)^j V_j$. Only K_0 and V_0 are singular (the Laplace kernels), and $K_1 = 0$. With $\tilde{p} = \Sigma_j (ik)^j \tilde{p}_j$, order j reads

$(\tfrac{1}{2}I - K_0)\,\tilde{p}_j = -V_j\,u_n + \Sigma_{m=2\ldots j}\, K_m\, \tilde{p}_{j-m}$.

One dense LU factorization of the real, frequency-independent Laplace operator $(\tfrac{1}{2}I - K_0)$ per shape serves every mode and every order; each order costs a back-substitution and products with smooth kernels. Open sheets use the hypersingular Laplace form for the pressure jump, expanded the same way.

```
factor A0 = (½I − K0) once per shape                  # dense LU, f64, CPU
for each mode n with k_n a < 0.5:
    p[0] = solve(A0, −V0 u_n)
    for j = 1, 2, …:  p[j] = solve(A0, −V_j u_n + Σ_{m=2..j} K_m p[j−m]);  stop when
the tail meets the target
    p_n = ρ0 c² k_n² Σ_j (i k_n)^j p[j];   a_lm = surface integral (§4.7), l ≤ 3
```

- **Orders and bound.** With t_j the size of the j-th term and $\rho = t_j/t_{j-1}$, the tail is estimated as $t_j\,\rho/(1 - \rho)$. On a sphere the series provably converges for $ka < 1$: the monopole response has the factor $1/h_0'(ka)$, and $h_0'(z) = e^{iz}(z + i)/z^2$ vanishes at $z = -i$, the nearest singularity (derived). Terms shrink about like $(ka)^j$, so at $ka = 0.5$ about seven orders reach the 1.4% solve target ($0.5^7 = 0.8\%$); six reach the 0.24 dB line ($0.5^6 = 1.6\%$).
- **Multipoles.** The truncation rule of §4.7 gives $N = 3.5$ at $ka = 0.5$ (three digits), so $l \leq 3$. The monopole is exact from the volume velocity $\int_\Gamma u_n\, dS$.
- **Size.** Panels follow the geometry and mode shapes, not the wavelength: about 1,000-5,000 unknowns, so $8N^2$ bytes for the dense f64 factor of N unknowns (8-200 MB). The band ends at 546 Hz for a 10 cm sphere-like object ($a = L/2$) and 315 Hz for a 10 cm cube (a its corner radius); 109 and 63 Hz at 50 cm (derived).

4.5 MW, the matched wave solve

One mode is one GPU run, driven at that mode's frequency only (Figure 3).

Grid. Spacing $h = \min(\lambda/n_{ppw}, L/40)$, with n_{ppw} = 6-8 points a wavelength; the floor $L/40$ keeps the shape resolved for low modes. A run's updates grow about as h^{-4}. A 5 kHz mode of a 50 cm object needs about 360 times fewer updates than a 29.4 kHz mode once the fixed 12-cell margins and the run length are counted (h^{-4} alone, $(8.6\text{ mm}/1.46\text{ mm})^4$, gives about 1,200; derived; Figure 2).

Domain. The object's bounding box, a margin of 2 air cells, then an absorbing layer 10 cells thick: the domain ends 12 cells off the box on each side. The grid is stored in bricks of 8^3 cells; bricks wholly deeper than 2 cells (the stencil's reach) inside the solid are neither stored nor updated.

Update. Leapfrog in time with the (2,4) stencil, $\nu = c\Delta t/h$:

$p^{n+1} = 2p^n - p^{n-1} + \nu^2 \Sigma_{d=1,2,3} (-p_{-2} + 16p_{-1} - 30p_0 + 16p_{+1} - p_{+2})/12$, five points along axis d.

It is stable for $\nu \le \tfrac{1}{2}$ in three dimensions; the method takes $\nu = 0.3$-0.5.

Dispersion, exactly. A plane wave e^(i(κ·x − ωt)) on the grid obeys

$\sin^2(\omega\Delta t/2) = \nu^2 \Sigma_d [s_d + s_d^2/3]$, $s_d = \sin^2(\kappa_d h/2)$,

which gives the phase-speed error ε in closed form for every direction (the scheme's phase errors are analysed in [37]). At (6 points, $\nu = 0.5$), (8, 0.5) and (8, 0.3), ε along an axis is +0.55%, +0.45% and +0.03%; the largest over all directions is 1.10%, 0.63% and 0.21% (derived; Table 1).

Prewarped drive (optional). In the steady state the leapfrog difference is exactly $-(4/\Delta t^2) \sin^2(\omega\Delta t/2)$. Driving at $\omega_d = (2/\Delta t) \arcsin(\omega\Delta t/2)$, with the boundary amplitude still set by ω, makes the steady state solve the space-discrete Helmholtz equation at the true k with no time error. Only the stencil's own error is left: 0.02-0.20% at 8 points a wavelength, 0.07-0.61% at 6, 0.01-0.08% at 10 (derived). A further small shift of ω_d centres this anisotropic error on zero (±0.09% at 8 points). The lock-in then reads at ω_d.

Absorbing layer. A perfectly matched layer [28] in a form for the second-order wave equation [29]; damping $\sigma(x) = \sigma_{max} (x/\delta)^m$ across its thickness δ, $m = 3$, $\sigma_{max} = (m + 1)\, c \ln(1/R)/(2\delta)$ for a nominal normal reflection $R = 10^{-4}$. Its real reflection is measured once per setting.

The surface on the grid. The ghost-fluid method [30] in its image-point form [31]. Every solid node within 2 cells of an air node is a ghost g. From the shape come its depth d, its closest surface point and the normal n. Its image point $x_I = g + 2d\, n$ lies in the air; $p(x_I)$ is interpolated from air nodes (trilinear, or least squares where neighbours are ghosts). The ghost value carries the drive:

$p_g = p(x_I) - 2d\, \rho_0 \omega^2 u_n\, R(t) \cos(\omega_d t)$, $R(t) = \tfrac{1}{2}(1 - \cos(\pi t/2T))$ for the first 2 periods ($T = 2\pi/\omega$), then 1.

This boundary is first to second order in h; refinement per kind of shape covers it. Open sheets thinner than h get two-sided ghosts, one set per face.

Run length and lock-in. The run lasts 3 domain crossings plus 4 periods, $3D/c + 4T$, with D the domain's side. The last 2 periods are lock-in windows: each node of the integration surface S accumulates $\Sigma\, p \cos(\omega_d t)$ and $\Sigma\, p \sin(\omega_d t)$, and one 2×2 least-squares solve, shared by all nodes, gives $p \approx \alpha \cos + \beta \sin$, so $P = \alpha + i\beta$. This is exact for any time step. The two windows give P_{prev} and P_{last}; the **leftover** $\|P_{last} - P_{prev}\|/\|P_{last}\|$ measures the transient still present. Above its budget (about 0.3%) the run continues for one more crossing.

Far field. S is a closed box halfway between the first and second air cells; P on S is the mean of the two cells and $\partial P/\partial n$ their difference over h. The coefficients come from the exact surface integral of §4.7 over S, so the far field needs no propagation through the grid. The surface pressure on Γ, for air loading, is read at the image points.

Side by side. Small runs (small objects, low modes) are launched together, up to 32 per launch, sharing one launch's overhead; each run keeps its own arrays, drive and lock-in.

Memory. About 12 bytes a cell (two time levels and a scratch array, f32).

```
h = min(λ/ppw, L/40);  dt = ν h / c;  ω_d = (2/dt)·asin(ω dt/2)          # prewarp
optional
grid = box + 2 air cells + 10 absorbing cells, in 8³ bricks; drop bricks > 2 cells
inside the solid
for step in 0 … (3D/c + 4T)/dt:
    ghost values from image points and the drive ρ0 ω² u_n R(t) cos(ω_d t)
    leapfrog update, (2,4) stencil; absorbing-layer update
    in each of the last two periods: accumulate p·cos(ω_d t), p·sin(ω_d t) on S
P_prev, P_last = least-squares amplitudes on S;  leftover = ‖P_last −
P_prev‖/‖P_last‖
a_lm = surface integral over S of (P_last, ∂P_last/∂n)
```

MW parameter	Default	Useful range
Points a wavelength; floor	6-8; L/40	5-14 (set by the certificate); L/20-L/80

MW parameter	Default	Useful range
$\nu = c\Delta t/h$	0.3-0.5	up to 0.5 (stability)
Air margin; absorbing layer	2 cells; 10 cells	2-10; 8-20
Run; lock-in windows	3 crossings + 4 periods; last 2 periods	2-6 crossings, 3-8 periods; 1-4 periods
Brick; runs per launch	8^3 cells; up to 32	4^3-16^3; 1-64

4.6 MB, the matched boundary solve

One mode is one frequency-domain solve on Γ only, at that mode's k (Figure 4).

Panels. Γ is triangulated at $h_p = \min(\lambda/n_p, L/60)$, n_p = 8-12 panels a wavelength [32], directly from the object's surface. Unknowns are piecewise-linear (P1), about one per h_p^2 of surface. Sheet-like parts much thinner than h_p become open sheets whose unknown is the pressure jump μ.

Equation. For closed parts, Burton-Miller's combined equation [15] with the coupling replaced by an OSRC operator [16, 17]:

$$(\tfrac{1}{2}I - K + M\,W)\,p = (M(\tfrac{1}{2}I + K') - V)\,q,\quad q = \rho_0\omega^2 u_n,$$

with V, K as in §4.4 at k, K′ the adjoint double layer, W the hypersingular operator $(Wp)(x) = \partial/\partial n_x \int_\Gamma \partial G/\partial n_y\, p\, dS_y$, and

$$M = (1/(ik))\,(1 + \Delta_\Gamma/k_\varepsilon^2)^{(-1/2)},\quad k_\varepsilon = k + i\varepsilon,\quad \varepsilon = 0.4\,k^{(1/3)}\,R^{(-2/3)},$$

where Δ_Γ is the Laplace-Beltrami operator and R the radius of the smallest sphere containing the object [17]. M approximates the exterior Neumann-to-Dirichlet map, so the left operator is close to the identity on smooth surfaces; M = 1/(ik) gives the classical Burton-Miller equation. W is used in Maue's regularized Galerkin form, for this sign of W: $\langle Wp, \varphi\rangle = -\int\int G(x,y)\,[\mathrm{curl}_\Gamma p(y)\cdot\mathrm{curl}_\Gamma \varphi(x) - k^2 (n_x\cdot n_y)\, p(y)\,\varphi(x)]\, dS_y\, dS_x$. Open sheets use $W\mu = q$, preconditioned by M on the left [18]; mixed objects solve both in one block system.

OSRC operator. The square root is localized by a rotated Padé approximation [33]: $\sqrt{(1+z)} \approx C_0 + \Sigma_{j=1\ldots N_P} A_j\, z/(1 + B_j\, z)$, with $a_j = (2/(2N_P+1))\,\sin^2(j\pi/(2N_P+1))$, $b_j = \cos^2(j\pi/(2N_P+1))$, $w = e^{(-i\theta)}$, $C_0 = e^{(i\theta/2)}\,[1 + \Sigma\, a_j(w-1)/(1 + b_j(w-1))]$, $A_j = e^{(-i\theta/2)}\, a_j/(1 + b_j(w-1))^2$, $B_j = w\, b_j/(1 + b_j(w-1))$. The method takes $N_P = 8$, $\theta = \pi/3$ [17]; these coefficients match $\sqrt{(1+z)}$ at sample points to a relative error of 10^{-9}-10^{-4}, worst near the branch cut (derived). With the surface finite-element mass matrix M_Γ and stiffness S_Γ, $Y = M\,X$ costs $N_P + 1$ sparse solves, all factored once per mode:

$(M_\Gamma - (B_j/k_\varepsilon^2)\, S_\Gamma)\, X_j = M_\Gamma\, X$ for each j; then $(M_\Gamma - S_\Gamma/k_\varepsilon^2)\, Y = (1/(ik))\,[C_0\, M_\Gamma\, X - \Sigma_j\, (A_j/k_\varepsilon^2)\, S_\Gamma\, X_j]$.

Products by FFT on a regular grid (pFFT [13]). A regular grid in the object's frame, spacing $h_g = \min(\lambda/n_g, L/30)$, n_g = 3.5-4, zero-padded to twice its size in each direction. One product: (1) **project** each panel's quadrature sources (monopoles for V and K′, dipoles for K and W) onto a stencil of 3^3-4^3 grid points that reproduces their potential beyond the stencil; (2) **convolve**: FFT, multiply by $\hat{G}$ (the FFT of G_k at grid offsets, once per mode), inverse FFT; (3) **interpolate** back to the panels' test functions (normal derivatives for K′ and W); (4) **precorrect**: for panel pairs whose stencils lie within about 2 grid cells, replace the grid's interaction by the exact one. As in [13], the projection weights reproduce each source's multipole moments about the stencil's centre up to the stencil's order (or its potential at test points around the stencil), and $\hat{G}$ at zero offset may take any finite value, since the precorrection removes the same grid interaction.

The exact near interactions split into a singular part, $G_0 = 1/(4\pi r)$, integrated once per shape with singular quadrature, and a smooth part, $G_k - G_0 = \Sigma_{j\ge 1} (ik)^j\, r^{j-1}/(4\pi\, j!)$, assembled per mode from moment integrals of r^{j-1} stored once per shape; 12-14 terms reach 10^{-4} over a near zone of half a wavelength ($\pi^{14}/14! \approx 10^{-4}$, derived). The multipole coefficients of §4.7 can be summed from the same grid sources.

Iteration. GMRES to a relative residual of 10^{-4}, with sums in a fixed order. Darbas et al. measured 4-7 iterations with this preconditioner on a sphere and a cone-sphere [17]; cavities need more. Their far-field

errors of 0.2-2.5% at 10 points a wavelength [17] suggest that 8 panels a wavelength may not hold 0.12 dB at the top band, hence the range up to 12.

Residual and certificate. After convergence one extra product gives the true residual $r = b - A\tilde{p}$, and $\|p - \tilde{p}\| \le \|A^{-1}\| \, \|r\|$. On a sphere every operator here is diagonal in spherical harmonics, so $\|A^{-1}\|$ is known in closed form from j_l and h_l [34]; elsewhere it is estimated as $1/\sigma_min$ of GMRES's own Hessenberg matrix (an estimate, not a proof). Each coefficient then obeys $|\Delta a_{lm}| \le k \, \|\partial_n(j_l \bar{Y}_{lm})\| \, \|\Delta p\|$, norms on Γ.

Memory. The FFT grids, Krylov basis and near-field corrections are freed when the mode is done.

```
panels at h_p = min(λ/n_p, L/60);  grid at h_g = min(λ/n_g, L/30), padded ×2;  Ĝ =
FFT of G_k on grid offsets
near = singular part (once per shape) + Σ_j (ik)^j/j! · moments_j;  osrc = factor N_P
+ 1 sparse systems at k_ε
b = M(½I + K')q − Vq
p, H = GMRES(A = ½I − K + M·W, b, tol 1e−4)      # product: project, FFT·Ĝ·IFFT,
interpolate, + near
r = b − A p;  certificate = ‖r‖·‖A⁻¹‖ (closed form on spheres, from H elsewhere);
a_lm = surface integral (§4.7)
```

MB parameter	Default	Useful range
Panels a wavelength; floor	8-12; L/60	6-15; L/30-L/120
FFT grid points a wavelength; floor; stencil	3.5-4; L/30; 3^3-4^3 points	3-6; 2^3-5^3
Padé terms; rotation; damping ε	8; π/3; 0.4 k^(1/3) R^(−2/3)	4-12; π/4-π/2; 0.2-1 times that
GMRES tolerance	10^{-4}	10^{-3}-10^{-6}

4.7 From the solution to the outputs

Multipole coefficients. From Green's representation and the addition theorem $G(x, y) = ik \sum h_l(k|x|) Y_{lm}(\hat{x}) j_l(k|y|) \bar{Y}_{lm}(\hat{y})$ ($|x| > |y|$, about x_0), over any closed surface S enclosing the object (Γ, or MW's box):

$$a_{lm} = ik \int_S [\, p \, \partial_n(j_l(k|y - x_0|) \bar{Y}_{lm}) - \partial_n p \cdot j_l(k|y - x_0|) \bar{Y}_{lm} \,] \, dS_y.$$

The order is $N = ka + 1.8 \, d^{(2/3)} \, (ka)^{(1/3)}$ with d = 3 digits [35]: at 29.4 kHz, 38-60 for a 10 cm object and 154-256 for 50 cm, depending on whether a is half the side or a cube's corner radius (derived). j_l, h_l and Y_{lm} come from stable recurrences in double precision.

Radiated power and damping. $P = \Sigma |a_{lm}|^2 / (2\rho o c k^2)$, taken from the far field, where every term is positive: for modes that radiate poorly the surface form $\tfrac{1}{2} \operatorname{Re} \int p \, \bar{v} \, dS$ cancels almost entirely. At unit modal amplitude the mode's energy is $\omega^2/2$, so radiation adds the decay rate $\sigma_rad = P/\omega^2$ to the material's σ_mat, and $T_{60} = 6.91/(\sigma_mat + \sigma_rad)$.

Air loading. $Z_{nn} = \int_\Gamma p_n \, u_n \, dS$ from the surface solution. The boundary data carry the factor ω^2, so write $Z_{nn} = \omega^2 \rho o B_{nn}$; the loaded frequency then solves $\omega^2 = \omega_n^2 + \omega^2 \rho o B_{nn}$ exactly: $\tilde{\omega} = \omega_n \, (1 - \rho o B_{nn})^{(-1/2)}$. Its real part is the pitch (lower for a mass-like load); its imaginary part is the radiation decay. To first order $\Delta\omega/\omega_n \approx \operatorname{Re} Z_{nn}/(2\omega_n^2)$; that shortcut errs by about $(3/8)(2\Delta\omega/\omega)^2$, 0.03% for a 1.5% shift (derived). The exact form leaves only B_{nn}'s own change across the shift: small at low ka, and handled by the second solve below where ka is large. As a check, $-(\omega/2) \operatorname{Im} Z_{nn}$ must equal the far-field power.

Near-equal modes. $Z_{mn} = \int_\Gamma p_m \, u_n \, dS$ comes from the same solves; reciprocity ($Z_{mn} = Z_{nm}$ at one frequency) is a further check. For modes closer than about $10 \, |Z_{mn}|/(2\omega)$, the small generalized eigenproblem $\Omega^2 \eta = \omega^2 (I - \rho o B) \eta$, with $B = Z/(\rho o \omega^2)$, gives the loaded frequencies, decay rates and mixed shapes; a mixed mode's coefficients are $\Sigma_n \, \eta_n \, a_{lm}^{(n)}$.

Second solve at the loaded pitch. The radiation pattern moves with ka; where $|\Delta\omega/\omega| \cdot ka$ exceeds about 0.1, the mode is solved again at its loaded frequency.

Accuracy targets. Each solve aims at 0.12 dB (1.4% in amplitude), half the 0.24 dB level step. Damping needs the radiated power to 10.7% (when radiation is 89% of the decay, for a 9.5% step); air loading needs the shift to 5.5% of itself (a 1.5% shift against a 0.083% step) (derived).

4.8 Certificates: how each error is bounded or checked

Part	Error source	Certificate	Proven or checked by
LB	The series' tail	Last term × ρ/(1 − ρ)	Proof on spheres (convergence for ka < 1); a-posteriori elsewhere; MW on the lowest modes
MW	Dispersion	Advisory: the phase-speed error per direction from the exact dispersion relation, and the phase φ it implies over the longest path from the surface to S, counted as 2 sin(φ/2)	Closed form (derived); zero time part with the prewarped drive. The operative check is refinement
MW	Steady state	The lock-in's leftover	Every run
MW	Absorbing layer; surface on the grid	Change at the next grid level	Refinement per kind of shape; spheres
MB	Iterative solve	$\|r\| \times \|A^{-1}\|$	Closed form on spheres; estimated from GMRES elsewhere
MB	Panels; projection; near-field series	Change at the next level	Refinement per kind of shape
All	Truncation; air loading and pairs	The order rule; second order in the shift	Standard [35]; a coupled structure-air solve

Combining, and the pass test. Each part is expressed as a relative error of the far-field coefficients, $\|\Delta a\|/\|a\|$, which by Parseval is the relative L2 error of the far field over directions. A phase error φ counts as 2 sin(φ/2); MB's surface error counts through the coefficient bound of §4.6. The stored certificate is the sum of the parts that apply (a deliberate over-estimate), with the largest named as its source, and a mode passes when the sum is within its budget (1.4%). The parts are: LB's tail; MW's lock-in leftover; MB's residual bound; the truncation (10^{-d}, 0.1% at d = 3); and the **refinement figure** for the shape's kind at the run's resolution and ka. That figure is the change measured when a reference run of that kind moves to the next level (points or panels a wavelength × r, r = 1.5), scaled by r^p/(r^p − 1) for the observed order p (Richardson: 1.8 for p = 2, 3 for p = 1). It covers what has no closed form: the boundary on the grid, the absorbing layer, dispersion along the real paths, panels and projection. Where no figure covers the kind and ka, the object's loudest mode in that band is also solved at the next level, and its change stands in.

The dispersion bound is advisory. It is stored beside the certificate but never blocks a result, because over long paths it far exceeds the real error. Over a path as long as the object, a 10 cm object at the top band allows 6.5° at 8 points and ν = 0.3, an 11% relative error, or 2.8° (5%) with the centred prewarp; a 50 cm object allows 32°, or 14° centred (derived). Most of the field on S comes from nearby surface over short paths: a few cells where the surface meets its box, up to about 0.37 L at the corners of a sphere's box. A kind whose refinement figure misses the budget runs with more points a wavelength (cost ∝ n_ppw^4) or goes to MB. No method proves an error bound in advance for an arbitrary shape; each result carries its own check.

Checks of the solvers. Pulsating, oscillating and higher-order spheres up to ka 135 (a 50 cm sphere at 29.4 kHz), each part within 0.12 dB a mode of the closed form, and radiation resistance and reactance within 1.4%; MW against a dense f64 CPU boundary solve on small objects; MB against MW; each refinement level against its certificate; LB's tail on spheres and against MW; reciprocal checks at a few listening points (§5).

4.9 Order of solves and the switch to exact results

```
on a new object:
    run LB for every mode with k a < 0.5                        # CPU, at once
    queue = the other modes, loudest first (by w_n)
    while queue is not empty:
        batch = next modes whose runs fit the GPU share and the free memory
        run MW or MB for each, each mode alone at its own ω
```

```
        for each result:
            no refinement figure for its kind and ka → also solve the loudest such
mode at the next level
            certificate (§4.8's sum) within budget → publish; the mode switches at
the next audio block boundary
            else → requeue with more points a wavelength, or with MB
            if |Δω/ω|·ka > 0.1 → requeue once at the loaded frequency
```

- Until its exact result lands, a mode plays an interim estimate, labelled as such (a radiation-efficiency formula for the shape's kind, or the nearest shape already solved). The mixer reads radiation data at block starts (for example 10 ms blocks), so each mode switches once, at a block boundary, and never back (Figure 5). A new shape is a new object.
- Solves run at a reserved GPU share (10-25%), never inside a frame's time budget, and only within the memory the frame leaves free. One solve's memory is freed when its mode is done, so more sounding objects never add memory: solves queue. Authored objects are solved ahead of time.

4.10 What runs where

Part	Processor	Precision	Working memory
LB	CPU, several cores	f64	$8N^2$ bytes for the factor of N unknowns, a shape
MW	GPU compute	f32 fields; f64 lock-in sums	About 12 B a cell
MB	GPU (FFTs, products, Krylov); CPU or GPU (sparse OSRC factors)	Complex f32 products; f64 reductions	FFT grids, Krylov basis, near-field corrections; freed per mode

Sums run in a fixed order, so results are bit-identical on one machine and differ across GPU makers by rounding only. Vendor-neutral GPU compute (for example Vulkan) lets the method run on any maker's GPU.

5. Variants

The method may also:

- **Split the bands differently:** any low-band edge below ka = 1; more bands; two parts only (LB with MW, or LB with MB); MW or MB chosen per mode from measured cost.
- **Do the low band differently:** expand the field instead of the surface data; Padé or other summation past ka = 1; Laplace problems by finite elements, fast multipole methods or on a GPU; measured or analytic modes.
- **Use other schemes in MW:** second-order (Yee-type) stencils at more points a wavelength; (2,6), (4,4) by the modified-equation method, or dispersion-relation-preserving and compact (Padé) stencils; staggered pressure-velocity grids; pseudo-spectral methods or adaptive rectangular decomposition; finite or spectral elements and discontinuous Galerkin; nested, octree or adaptive grids; one grid reused for modes of nearby frequency, each still driven and read alone; storage in f16, f32 or f64.
- **Absorb differently:** split-field, convolutional or complex-frequency-shifted PMLs; sponge layers; Engquist-Majda or Higdon conditions; exact non-reflecting boundaries; S anywhere between object and absorber, as a box, a sphere or a surface following the object.
- **Reach the steady state differently:** other windows or fits; subtracting a fitted decaying transient; controllability or WaveHoltz iterations [23, 24] with Krylov acceleration; warm starts from a solved neighbour mode; a broadband drive of one mode's shape with Fourier extraction at several frequencies (for the loaded pitch).
- **Treat the surface differently in MW:** cut cells, immersed-interface or embedded-boundary methods, conforming or curvilinear grids, sharp-interface boundary operators, a finer grid near the surface only.
- **Use other boundary equations in MB:** CHIEF; indirect single- or double-layer forms; Brakhage-Werner; generalized combined equations with square-root operators [16]; Calderón preconditioning; collocation or Nyström discretizations; curved or higher-order elements;

isogeometric or time-domain BEM; a frequency-domain volume solve on the matched grid with a PML and a shifted-Laplacian, sweeping or domain-decomposition preconditioner.

- **Precondition and iterate differently:** near-field sparse approximate inverses, block-diagonal or ILU preconditioners, multigrid, other OSRC forms or Padé orders; Krylov subspaces recycled across modes of nearby frequency [36]; restarted or flexible GMRES, BiCGStab, COCG or QMR, block Krylov methods for several modes at once.
- **Use other fast products:** single- or multilevel and broadband fast multipole methods [10]; H- or H^2-matrices with adaptive cross approximation [11]; IFGF [12]; AIM [14]; butterfly or non-uniform FFT methods; the near field computed directly per mode or interpolated across modes in frequency.
- **Keep other runtime forms:** multipoles at reduced order where listening tests allow; equivalent sources [19] fitted to the exact solve on a check surface just off the object, valid closer than the multipole sphere; directivity tables with phase; per-band powers for diffuse propagation; rotated (Wigner-D) or translated multipoles for moving objects.
- **Run reciprocally:** a point source at each of a few listening points gives, in one run per point, every mode's pressure there (each mode's surface velocity integrated against the reciprocal field at its frequency); as a check, or as the method when a few fixed points matter (measurement microphones, a fixed seat).
- **Certify differently:** a refined second solve (Richardson); goal-oriented adjoint estimates for the coefficients; other residual norms; closed-form shapes as calibration for kinds of shapes.
- **Schedule differently:** by frequency, contact point, distance to the listener or audibility after masking; cross-fades over one or more blocks; the power first and the directivity later.
- **Share and run elsewhere:** key results by a shape digest and share them between copies, machines or a network; bake them offline; re-solve only modes a shape change affects; run on any GPU through Vulkan or another compute interface, integrated GPUs, CPUs with vector units, standalone headsets, clusters, rented cloud machines, DSPs or FPGAs.
- **Couple more physics:** objects in contact radiating as one surface; a reflecting floor (a half-space Green's function) or a room (its Green's function, or a grid that includes it); water and other heavy fluids, where a coupled structure-fluid solve replaces the first-order loading rule.
- **Serve other fields:** film and animation sound; product and vehicle acoustics (appliances, engines, gearboxes, electric motors, tyres); loudspeaker and instrument design (cones, enclosures, bells, plates, guitar bodies); underwater acoustics and sonar; medical and industrial ultrasound; architectural acoustics (panels and partitions); non-destructive testing; robotics (contact sounds for perception); certified training data for learned models; and electromagnetics, where each resonance of an antenna, cavity or metamaterial resonator is solved alone at its frequency by FDTD on a matched grid or by a method of moments with FFT products and Calderón or OSRC-type preconditioning.

6. Worked example

Each value is derived from the formulas of §4.

The bands. LB ends at ka = 0.5: 546 Hz for a 10 cm sphere-like object (a = L/2 = 5 cm), 315 Hz for a 10 cm cube (a its corner radius, 8.7 cm), and 109 and 63 Hz at 50 cm. At 29.4 kHz the same objects are at ka 27, 47, 135 and 233.

MW's grid. For a 50 cm object at 8 points a wavelength, a 5 kHz mode (λ = 68.6 mm) runs on h = 8.6 mm cells, 58 across the object, and a 29.4 kHz mode (λ = 11.7 mm) on 1.46 mm cells, 343 across; the floor L/40 = 12.5 mm does not bind. At ν = 0.3 the time steps are 7.5 μs and 1.28 μs.

MB's panels. At 29.4 kHz, panels of $\lambda/8$-$\lambda/12$ (1.46-0.97 mm), one unknown per h_p^2 of surface, put 28,000-64,000 unknowns on a 10 cm cube and 0.7-1.6 million on a 50 cm cube.

Table 1. Phase-speed error of the (2,4) leapfrog scheme (derived).

Points a wavelength, ν	Along an axis	Largest over all directions	With the prewarped drive (waves slow)
6, 0.5	+0.55%	1.10%	0.07-0.61%
8, 0.5	+0.45%	0.63%	0.02-0.20%
8, 0.3	+0.03%	0.21%	0.02-0.20%
10, 0.3	+0.07%	0.14%	0.01-0.08%
12, 0.3	+0.06%	0.10%	0.005-0.04%

For comparison, a second-order scheme at 7 points a wavelength and $\nu = 1/\sqrt{3}$ is 2.3% slow along an axis: 8° of phase per wavelength travelled (derived).

What is kept. Complex f32 multipoles at full order, $d = 3$, $8(N + 1)^2$ bytes a mode: at 29.4 kHz 0.01-0.03 MB (10 cm) and 0.19-0.53 MB (50 cm). With modes spread from each object's lowest mode to 29.4 kHz, per object 0.2-0.4 MB (a 10 cm solid, 20 modes from 14.6 kHz), 1.5-3.4 MB (a 10 cm shell, 292 modes from 0.3 kHz), 60-164 MB (a 50 cm solid, 500 modes from 1 kHz) and 139-374 MB (a 50 cm shell, 2,000 modes from 0.1 kHz). Evaluating 2,000 modes at full order for two ears 100 times a second costs 5-12 GFLOP/s for 10 cm objects and 77-212 GFLOP/s for 50 cm objects, so above $ka \approx 50$ the runtime form is a cost of its own for any method; the reduced forms of §5 address it.

7. Figures



Figure 1. Which part solves which mode. Modes are sorted by ka. Below 0.5, LB solves every mode at once on the CPU; its series converges on a sphere up to ka = 1. Above, each mode is one GPU run (MW) or one boundary solve (MB, for large objects). For sphere-like objects (a = L/2) the band edge is 546 Hz at 10 cm and 109 Hz at 50 cm, and 29.4 kHz is ka 27 and ka 135. For cubes, a is the corner radius: 315 Hz and ka 47 at 10 cm, 63 Hz and ka 233 at 50 cm (derived).

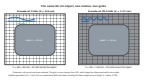


Figure 2. A grid matched to each mode (schematic). For a 50 cm object, a 5 kHz mode runs on 8.6 mm cells (58 across) and a 29.4 kHz mode on 1.46 mm cells (343 across); cell counts are drawn reduced. The 5 kHz run needs about 360 times fewer updates once the fixed margins and the run length are counted (h^{-4} alone: about 1,200; derived).

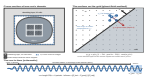


Figure 3. One MW run. Top left: the domain, with bricks deep inside the object skipped, the integration surface S in a 2-cell air margin and a 10-cell absorbing layer. Top right: a ghost node g takes its value from its image point x_I and the mode's drive. Bottom: the drive ramps up over 2 periods, the field settles over 3 crossings, and the last 2 periods are read by lock-in; their difference is the leftover.

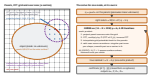


Figure 4. One MB solve. Left: panels on the outer surface, the regular FFT grid, one panel's projection stencil and its near zone, where exact interactions replace the grid's. Right: the right-hand side, GMRES with products by projection, FFT and interpolation, the true residual from one extra product, the certificate and the outputs.



Figure 5. Order of solves for a new object. The low band is solved at once on the CPU; the other modes are solved on the GPU loudest first, small runs side by side. Each mode plays a labelled interim estimate until its exact result lands, then switches once, at an audio block boundary.

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